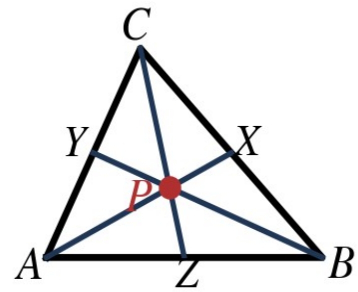
SC Honors Geometry - Nappier
6.3 – Medians and Altitudes of Triangles • Homework

Name: ___________________________________

In the figure at right, P is the centroid of △ABC.



1. Explain the difference between a median and an altitude of a triangle.

2. AP = 14. Find PX and AX.

3. BP = 5x and PY = x + 6. Find x and BY.

4. CZ = 36. Find CP and PZ.

Find the centroid of the triangle with the given vertices.

5. A(−2, 4), B(4, 6), C(4, −4)

6. D(1, 7), E(−5, −2), F(7, 1)

Answer each question about altitudes and the orthocenter.

7. Where is the orthocenter of a right triangle?

8. Where is the orthocenter of an obtuse triangle?

9. Find the orthocenter of the triangle with vertices (0, 0), (6, 0), and (0, 4).

10. A triangular metal plate must balance on the tip of a single post. Which point of concurrency should the post touch, and why?